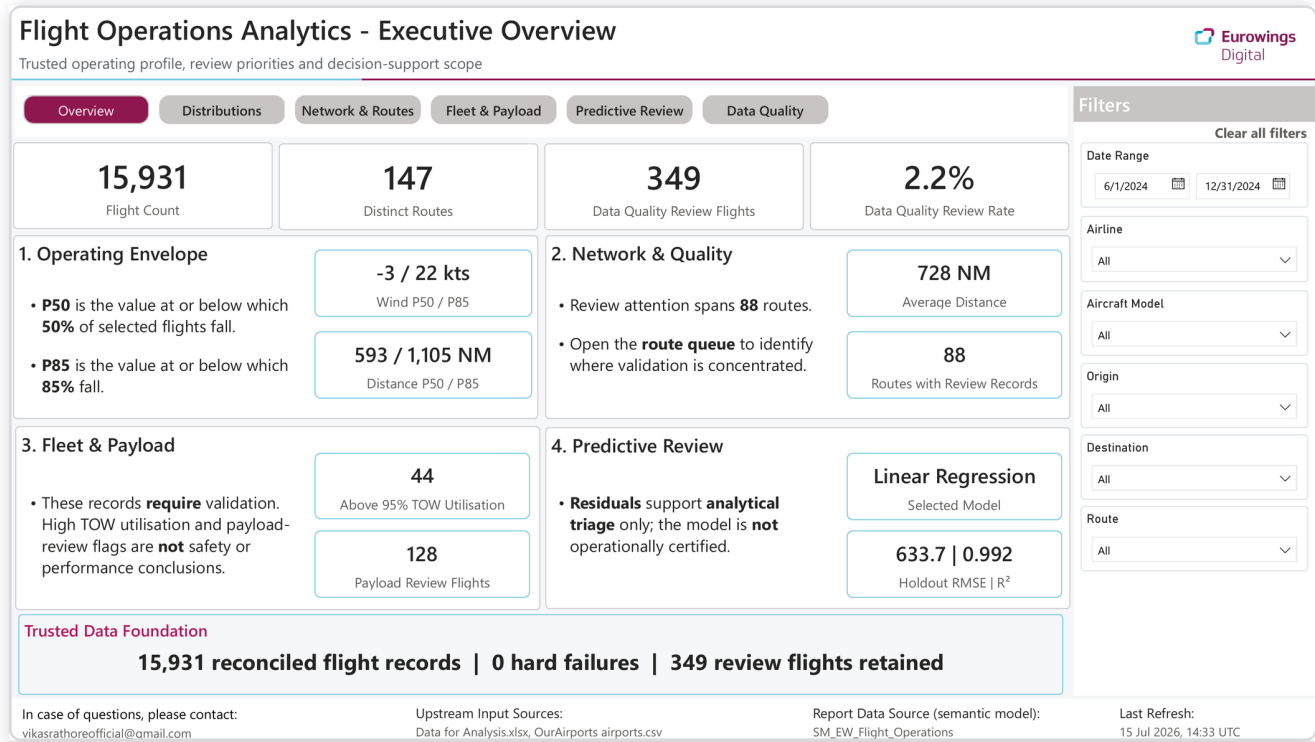


End-to-End Use Case Report

Eurowings Digital Business Analyst Case Study



0. Executive Summary

A governed analytical data product built from the supplied case-study dataset

Use case in one sentence

I converted a flight-level Power BI distribution task into a reproducible Microsoft Fabric and Power BI solution that moves from source ingestion to governed data, analytical features, a Trusted semantic model and an action-oriented report.

The mandatory requirement was to show wind and distance distributions, cumulative coverage and 50th/85th percentile reference points. I kept those outputs central, then extended the solution with data quality, route prioritisation, fleet and payload analysis, and a planned-TOW predictive prototype. The extension was designed to demonstrate how I would translate an operational question into a scalable data product without overstating what the available data can prove.

Validated outcome

15,931 Source and Trusted flights	147 Distinct routes	349 Review flights retained	0 Hard failures
-3 / 22 kts Wind P50 / P85	593 / 1,105 NM Distance P50 / P85	44 High TOW review flights	128 Payload review flights

What I delivered

- A layered Lakehouse implementation using Raw, Standardized, Curated and Trusted schemas.
- An end-to-end Fabric pipeline with parallel source branches, validation and explicit failure handling.
- A Direct Lake star-schema semantic model with governed DAX measures and no report-local composite model.
- A six-page Power BI decision product covering the mandatory task and the additional operational analyses.
- A chronological machine-learning holdout comparing Linear Regression and Gradient-Boosted Trees.
- Transparent assumptions and limitations, including unconfirmed weight units and unconfirmed signed-wind semantics.

Decision principle

Every analytical page follows the same flow: signal -> diagnosis -> underlying records -> defensible interpretation -> next action or required validation.

1. Business Requirement and Scope

Traceability from the case-study brief to the implemented solution

The operations team wanted to understand how flights are distributed across wind speed and travelled distance, where most flights fall, and how cumulative coverage reaches the median and upper coverage benchmark. The brief also invited additional insights, external data recommendations and Python or R analysis.

Requirement-to-solution traceability

Brief requirement	Implementation	Evidence
Wind distribution	10-knot wind bands with flight-count columns.	Page 01 - Distributions
Distance distribution	100-NM distance bands with flight-count columns.	Page 01 - Distributions
Cumulative percentage	Dynamic cumulative lines calculated in the semantic model.	Both mandatory charts
50th percentile	Global P50 reference and selected-scope P50 card.	-3 kts and 593 NM
85th percentile	Global P85 reference and selected-scope P85 card.	22 kts and 1,105 NM
Clear interpretation	Record list and decision guide below the charts.	Band selection filters exact flights
Additional insights	Route, fleet, payload, data-quality and predictive pages.	Pages 02-05
Python or R exploration	PySpark/Python notebooks for profiling, rules, aggregates and ML.	NB01-NB06

Scope boundary

The source supports descriptive operating profiles, data validation and analytical triage. It does not contain the operational outcomes required to conclude delay causality, fuel burn, safety risk, profitability or market share. I therefore separated supported analysis from unsupported claims throughout the pipeline, semantic model and report.

Success criteria used

- All source rows reconcile through the governed pipeline unless a documented hard-failure rule excludes them.
- Stable logic is implemented once upstream; dynamic filter-context calculations remain in DAX.
- Each KPI has a defined population, grain and limitation.
- Aggregated findings can be traced to underlying flight records.
- Model outputs are presented as analytical prototypes rather than operational certification.

2. Data Understanding and Assumptions

Source structure, reference enrichment and interpretation controls

Primary flight source

The supplied workbook contains 15,931 flight-level records across June-December 2024 and 11 business columns. One row represents one flight record.

Source column	Governed name	Business meaning
Flight Date	flight_date	Flight calendar date
Airline	airline_code	Operating-airline code
PIn Payload	planned_payload_weight	Supplied planned payload
PIn TOW	planned_tow_weight	Supplied planned take-off weight
PAX	passenger_count	Passenger count
Acft MTOW	aircraft_mtow_weight	Aircraft maximum take-off weight
Acft Type	aircraft_type_code	Aircraft type code
Origin	origin_iata_code	Origin IATA code
Destination	destination_iata_code	Destination IATA code
Airway Distance Travelled (Nautical miles)	airway_distance_nm	Travelled airway distance
Wind in Kts	wind_component_kts	Signed wind-component field

Reference enrichment

I added the OurAirports airport reference as an independent source branch. The complete 85,720-row file is retained in Raw and Standardized, while Curated restricts the reference to airports used by the flight dataset. This enables governed airport names, geography and role-playing origin/destination dimensions.

Critical assumptions

Weight unit

The dataset does not explicitly confirm the unit for payload, planned TOW and MTOW. Technical fields therefore use the neutral suffix _weight. The report does not relabel these values as kilograms or tonnes.

Signed wind semantics

The source definition does not confirm whether negative and positive values map to headwind or tailwind. The report uses the business-safe label Wind Component and avoids directional operational conclusions.

3. Solution Architecture and Orchestration

A small medallion-style data product implemented in Microsoft Fabric

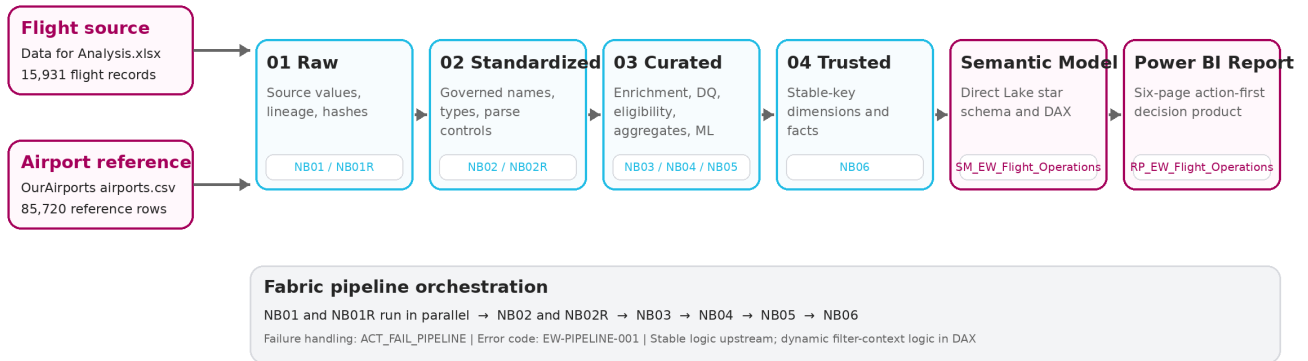


Figure 1. End-to-end lineage from landing files to the Direct Lake Power BI report.

Architecture decisions

- Raw preserves received values, source order, hashes and technical lineage; no business cleansing occurs there.
- Standardized applies governed names, parsing, normalization and source-to-target reconciliation.
- Curated contains business enrichment, data-quality rules, analytical eligibility, aggregates and ML outputs.
- Trusted publishes stable surrogate-key dimensions and facts for Direct Lake consumption.
- The flight and airport branches run independently until Curated, allowing parallel ingestion and standardization.
- The pipeline executes eight notebook activities and invokes ACT_FAIL_PIPELINE with error code EW-PIPELINE-001 if the final publication is skipped or fails.

Calculation placement

Heavy, stable and reusable logic is implemented in Fabric notebooks. Dynamic calculations that depend on the report filter context are implemented as DAX measures. Presentation text and interaction remain in Power BI.

4. Notebook Implementation

Clear responsibilities, validated outputs and reproducible execution

Notebook	Responsibility	Main output	Validation result
NB01	Ingest raw flight source	01raw.flight_raw	15,931 rows; lineage and hashes; succeeded
NB01R	Ingest airport reference	01raw.airport_reference_raw	85,720 rows; independent branch; succeeded
NB02	Standardize flight data	02standardized.flight_standardized	15,931 rows; 0 parse errors; reconciled
NB02R	Standardize airport reference	02standardized.airport_reference_standardized	85,720 rows; 0 parse errors
NB03	Curate quality and features	flight_enriched, DQ catalog/results, references	15,931 rows; 349 review flights; 0 hard failures
NB04	Build analytical products	distributions, percentiles, route/fleet aggregates	Global P50/P85 and reusable analysis tables
NB05	Train and evaluate models	scoring, metrics, feature importance, run summary	13,984 train; 1,819 holdout; LR and GBT
NB06	Publish Trusted model	11 dimensions and 11 facts	22 Direct Lake-ready tables; validated publication

Implementation conventions

- snake_case physical names and business-readable measure names;
- stable surrogate keys in Trusted and single-direction dimension-to-fact relationships;
- role-playing origin and destination airport dimensions;
- technical and lineage fields hidden from business users unless required for grain or validation;
- DIVIDE and COALESCE in DAX where denominator or blank handling matters;
- explicit model-run metadata, source reconciliation and validation assertions in the notebooks.

Resolved engineering issues

During development I corrected source-column renaming, Fabric Lakehouse context, airport-reference schema drift, notebook cell execution dependencies, pipeline timeout formatting, Trusted date coverage, relationship cardinalities and Direct Lake schema synchronization. These fixes are reflected in the submitted notebook exports and final model.

5. Data Quality and Analytical Eligibility

Retain review records, exclude only hard failures and define population by use case

Treatment rule

Hard failures are excluded from Trusted. Review warnings are retained and surfaced. Eligibility is evaluated separately for distributions, route analysis, weight, payload, carrier comparison and model training.

Review-rule results

Rule	Review condition	Trigger rows
DQ101	Zero planned payload with positive passenger count	128
DQ102	Origin equals destination	5
DQ103	Absolute wind component above 60 kts	175
DQ104	Absolute wind component above 80 kts	5
DQ105	TOW utilisation above 95%	44
DQ106	Legacy airline code	1

The 358 rule-trigger rows map to 349 distinct review flights, so triggers and flights are not interchangeable. DQ104 is also a subset of DQ103. The overall review rate is 2.2%, and review records are present across 88 routes.

Analysis-specific eligibility

Population	Eligibility principle
Wind distribution	Valid governed wind field and no applicable hard failure.
Distance distribution	Positive governed distance and no applicable hard failure.
Route analysis	Governed route-analysis flag; not inferred from visual filters.
Weight analysis	Planned TOW and MTOW positive, with planned TOW not above MTOW.
Payload analysis	Positive payload not above planned TOW, with positive passenger count.
Model training	Complete target/features, positive payload and no hard failure.

Derived operational features

The curated layer calculates TOW utilisation, remaining MTOW margin, payload-to-TOW ratio, payload-to-MTOW ratio, non-payload planned mass and a planned-payload-per-passenger proxy. The proxy is explicitly not presented as average passenger weight.

6. Analytical Products and Predictive Prototype

Reusable descriptive outputs plus a controlled planned-TOW model comparison

Curated analytical products

- Global wind and distance distributions with transparent bins and percentile references.
- Route profiles, route-month profiles, carrier profiles and aircraft-route deployment profiles.
- Route-distance variation and outlier features for investigation rather than automatic classification.
- Analysis-population summaries and metric correlations for trust and model review.
- Record-level model scoring, model-level metrics and GBT logical feature importance.

Predictive design

Design element	Implementation
Target	Supplied planned TOW
Split	Chronological last-calendar-month holdout
Training population	13,984 flights, June-November 2024
Test population	1,819 flights, December 2024
Numeric features	Payload, distance, wind, passenger count, MTOW, month
Categorical features	Aircraft type, route and airline
Models	Linear Regression and Gradient-Boosted Tree Regression
Selected model	Linear Regression

Holdout performance

Model	MAE	RMSE	R-squared	Role
Linear Regression	398.2	633.7	0.992	Selected
Gradient-Boosted Tree	732.4	1,061.3	0.979	Comparison

Interpretation boundary

Residuals are used to prioritise records for investigation. The model is not a dispatch, safety or certification model. Published model metrics are holdout-level metrics and must not be presented as route-specific recalculations.

7. Trusted Semantic Model and DAX Layer

A central Direct Lake model designed for reuse, filter consistency and clear grain

NB06 publishes a Trusted star schema to SM_EW_Flight_Operations. The Power BI report remains live-connected to the central Direct Lake model; I did not add a local report model or duplicate upstream transformations in Power BI.

Trusted tables

Group	Published tables
Dimensions	dim_date, dim_airline, dim_aircraft, dim_origin_airport, dim_destination_airport, dim_route, dim_wind_band, dim_distance_band, dim_dq_rule, dim_analysis_population, dim_model
Core facts	fact_flight, fact_dq_rule_trigger, fact_route_month, fact_aircraft_route
Predictive facts	fact_model_scoring, fact_model_metric, fact_model_feature_importance
Distribution and control facts	fact_percentile_reference, fact_analysis_population, fact_wind_distribution_global, fact_distance_distribution_global

Relationship principles

- Active, single-direction relationships from dimensions to facts.
- No fact-to-fact relationships.
- Stable surrogate keys and a marked date dimension.
- Separate origin and destination airport dimensions to prevent ambiguous filtering.
- dim_model filters scoring, metrics and feature-importance facts.
- Common date, airline, aircraft, route, origin and destination slicers filter the scoring fact.

Governed measure layer

Measures are organised into Core, Distributions, Percentiles, Route and Network, Aircraft and Weight, Payload and Passenger, Data Quality, Predictive Analytics and Executive Overview folders. Numeric base measures remain numeric; unit-bearing display measures are separate. Global benchmarks use controlled filter removal, while selected-scope percentiles respond to report filters.

Grain rule

A field belongs to the table whose grain it describes: flight classifications in fact_flight, route-month metrics in fact_route_month, aircraft-route metrics in fact_aircraft_route, model scores in fact_model_scoring and model metrics in fact_model_metric.

8. Power BI Decision Product

Six visible pages built around signal, diagnosis, records and action

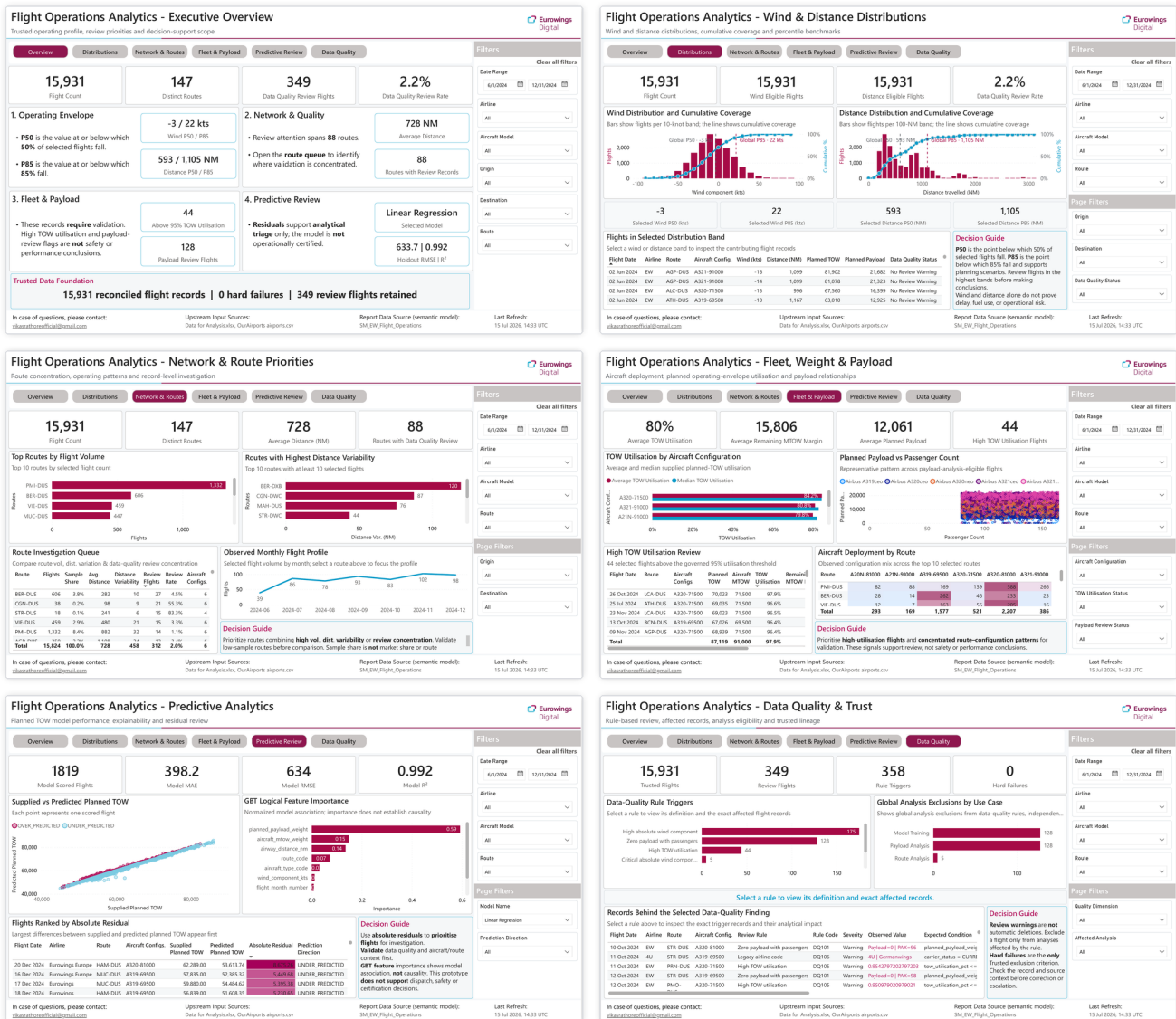


Figure 2. Final report pages: Overview, Distributions, Network & Routes, Fleet & Payload, Predictive Review and Data Quality.

Page	Decision purpose
Overview	Summarises the operating envelope, network quality, fleet/payload review, selected model and Trusted foundation.
Distributions	Delivers the mandatory wind and distance distributions, cumulative coverage, P50/P85 and record list.
Network & Routes	Prioritises routes using volume, distance variability, review concentration and observed monthly profile.
Fleet & Payload	Reviews TOW utilisation, payload versus passenger count, route deployment and high-utilisation records.
Predictive Review	Compares supplied and predicted planned TOW, model metrics, GBT importance and residual-ranked flights.
Data Quality	Shows rule triggers, analysis exclusions and exact affected records for correction or escalation.

Interaction contract

Shared slicers control date, airline, aircraft, route, origin and destination. Page-specific slicers add governed context such as data-quality status, fleet review status, model and prediction direction. Chart selections filter the relevant record or priority table; reset bookmarks restore the intended default state. Model-level metrics remain fixed to the selected published model and are not falsely recalculated for route subsets.

9. Key Findings and Business Interpretation

The main evidence produced by the final solution

Operating envelope

Half of flights fall at or below -3 kts for the supplied wind component and 593 NM for distance. Eighty-five percent fall at or below 22 kts and 1,105 NM. These are coverage benchmarks, not risk thresholds.

Network profile

The sample contains 147 routes with an average distance of 728 NM. Review attention spans 88 routes, so route prioritisation should consider review concentration and distance variability alongside volume.

Data quality

All 15,931 records reconcile to Trusted with zero hard failures. There are 358 review triggers across 349 distinct flights, equal to a 2.2% review rate. Warnings are retained for transparent analysis.

Fleet and payload

Forty-four eligible flights exceed 95% TOW utilisation and 128 flights have zero planned payload despite positive passenger count. Both are validation conditions, not evidence of unsafe or poor performance.

Predictive review

Linear Regression performs better than the GBT comparison on the December holdout, with RMSE 633.7 and R-squared 0.992. Residuals provide a ranked review queue but do not establish operational suitability or causality.

How the findings should be used

- Use P50 to describe the typical selected operating population and P85 to define an upper coverage boundary for scenario review.
- Investigate tail-band, route-variability, high-utilisation, payload-review and large-residual records in their exact flight context.
- Validate source definitions and additional operational outcomes before converting descriptive associations into operational decisions.

10. Additional Data, Limitations and Production Roadmap

What would convert the prototype into a stronger operational data product

Recommended additional data

Domain	Additional fields	Decision value
Operations	Flight number and leg ID, aircraft registration, scheduled/actual times, delay codes, actual route, flight time, altitude profile and detailed weather.	Delay analysis, tail-level patterns, operational reliability and weather-adjusted interpretation.
Fuel	Block, taxi, trip, reserve and landing fuel, plus uplift.	Fuel-efficiency analysis, route/aircraft comparison and planning variance.
Capacity and payload	Sellable seats, cabin configuration, baggage, cargo, mail, volume, MZFW, MLW and tail-specific DOW/OEW.	True load factor, cargo analysis, payload constraints and aircraft-tail context.
Commercial	Booking and flown revenue, fare family, channel, point of sale, ancillary revenue, no-show rate, charges and handling cost.	Route economics, customer and sales analysis, profitability and distribution decisions.

Current limitations

- Weight units and signed-wind semantics are not confirmed by the source.
- The sample covers June-December 2024 and should not be treated as full-year seasonality or market share.
- The source does not contain delay, fuel, safety, cost or revenue outcomes.
- High TOW utilisation, route-distance variation and model residuals are review signals, not operational failure labels.
- Feature importance represents association within the fitted GBT model and does not establish causality.
- The model predicts supplied planned TOW, not actual TOW.

Production hardening roadmap

1. Parameterise source locations and environment configuration; add run-control logging and alerting.
2. Confirm source definitions, units and data ownership; formalise acceptance thresholds with operations stakeholders.
3. Introduce deployment pipelines, access control, refresh monitoring and certified semantic-model governance.
4. Add actual operational outcomes and conduct drift, bias, stability and model-monitoring reviews before operational use.
5. Implement a governed residual-band or anomaly policy only after the business defines the threshold and action.

11. Conclusion and Submission Package

How the use case demonstrates end-to-end Business Analyst ownership

I treated the brief as both an analytical requirement and a data-product design exercise. The solution fulfils every mandatory Power BI requirement while showing how I would define logic, challenge unsupported interpretations, translate requirements into engineering work, publish a reusable semantic model and give operational users a clear path from signal to underlying record.

Role-to-evidence mapping

Business Analyst capability	Evidence in this submission
Translate business requirements	Requirement traceability, population definitions and decision boundaries.
Design data products and workflows	Layered Lakehouse, parallel source branches, notebook responsibilities and pipeline dependencies.
Define KPIs and transformation logic	Governed features, eligibility flags, percentile logic and DAX measure folders.
Bridge business and engineering	Business-safe naming, source caveats, stable keys, grain rules and Direct Lake synchronization.
Deliver analytics and AI capabilities	Six-page Power BI report plus chronological LR/GBT planned-TOW prototype.
Communicate clearly	Decision guides, record-level evidence, limitations and a concise executive overview.

Submission package

Artifact	Purpose
Use Case Report	This document - end-to-end business and technical narrative.
Notebook exports	NB01, NB01R, NB02, NB02R, NB03, NB04, NB05 and NB06 HTML exports.
Pipeline definition	PL_EW_Flight_Operations_EndToEnd.json.
Power BI file	RP_EW_Flight_Operations.pbix.
Power BI PDF	RP_EW_Flight_Operations.pdf with the six report pages.
Visual documentation	Data_Pipeline_Snapshot.png, Fabrics_Workspace_Snapshot.png, Semantic_Model_Snapshot.png with pipeline, fabrics workspace and semantic-model screenshots.

Final statement

The final solution is a reproducible, governed and explainable analytical prototype. It keeps the mandatory distributions at the centre, adds decision-relevant operational context, and remains explicit about what the available data can and cannot support.

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